

# Level Design

Introduction to Games

# hi!

I'm Alina.

Visual Design – Game Design – Humanities

 [alina.menten@fh-hagenberg.at](mailto:alina.menten@fh-hagenberg.at)

 FH1.234 (FH1 Ebene 2)

 Alina Menten



# Today's Lecture:

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## **Level Design**

Level Design vs. Game Design vs. Environment Art

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## **Level Elements**

Challenges, Motivation, Skills & Player Experience

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## **Spatial Composition & Building Vocabulary**

Types of Levels, Affordances & Readability

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## **How to make a level**

From Ideation to Playtesting

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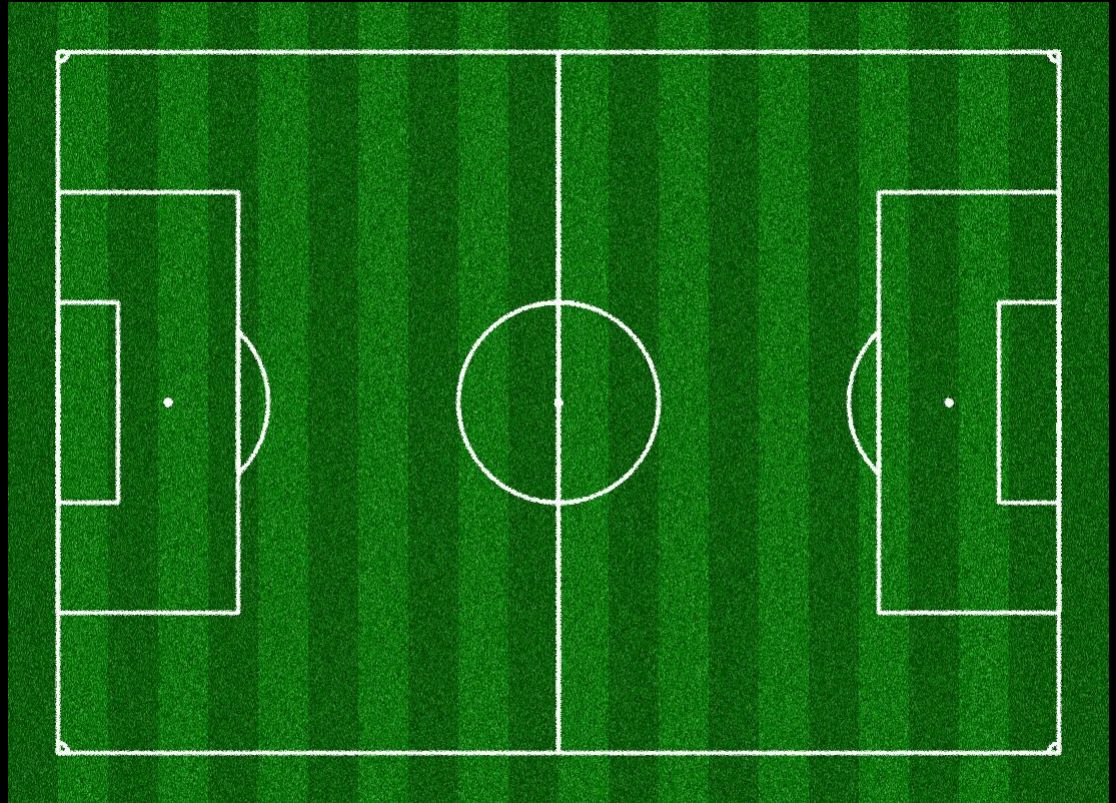
# Level Design.



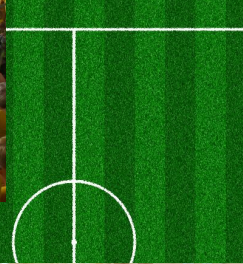
# Level Design = Design of Game Spaces

“Spaces set  
boundaries for  
players to move  
and interact.”

The Level Design Book







“Level designers, or map designers,  
are the individuals responsible for **constructing  
the game spaces** in which the **player competes**.

As such, the level designer is largely responsible  
for the implementation of the **game play  
in a title.**”

Sam Shahrani  
(Researcher)

**Level Design**  
**=**  
**Game Design?**



# Game Designer

- Designs and formulates rules  
(Whole Game + Gameplay)
- Designs Systems



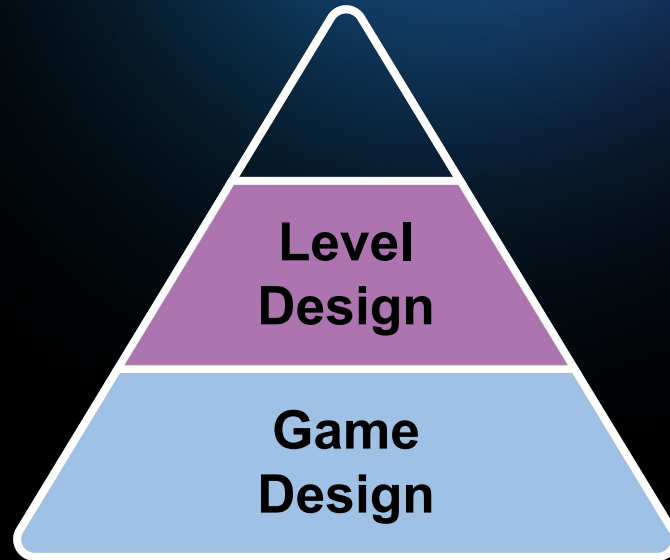


**“A level designer  
translates systems into  
experience through space.”**

The Level Design Book

# Level Designer

- Designs the player's encounter with rules
- Interprets rules into play
- Turns systems into experience.





**“A level designer has a  
very responsible position,  
because maps are where the  
game takes place.”**

John Romero  
(Commander Keen, Doom, Quake)

# Game Designer

What? + Why?

Whole game system

Rules

Mechanics, system goals

Invents grammar  
Represents theory (thinks in abstraction)

# Level Designer

How? + Where?

Specific playable spaces

Spatial experiences

Transforms abstract systems into felt  
gameplay experiences (Layouts, Maps,  
Wayfinding, Blockouts, ...)

Writes sentences based on grammar  
Represents practice (acts as  
translator/interpreter)

**ASKS**

**SCOPE**

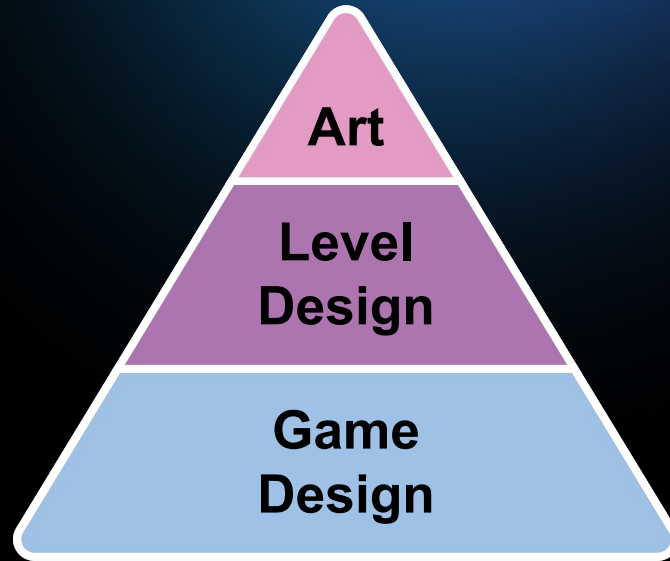
**FOCUS**

**OUTPUT**

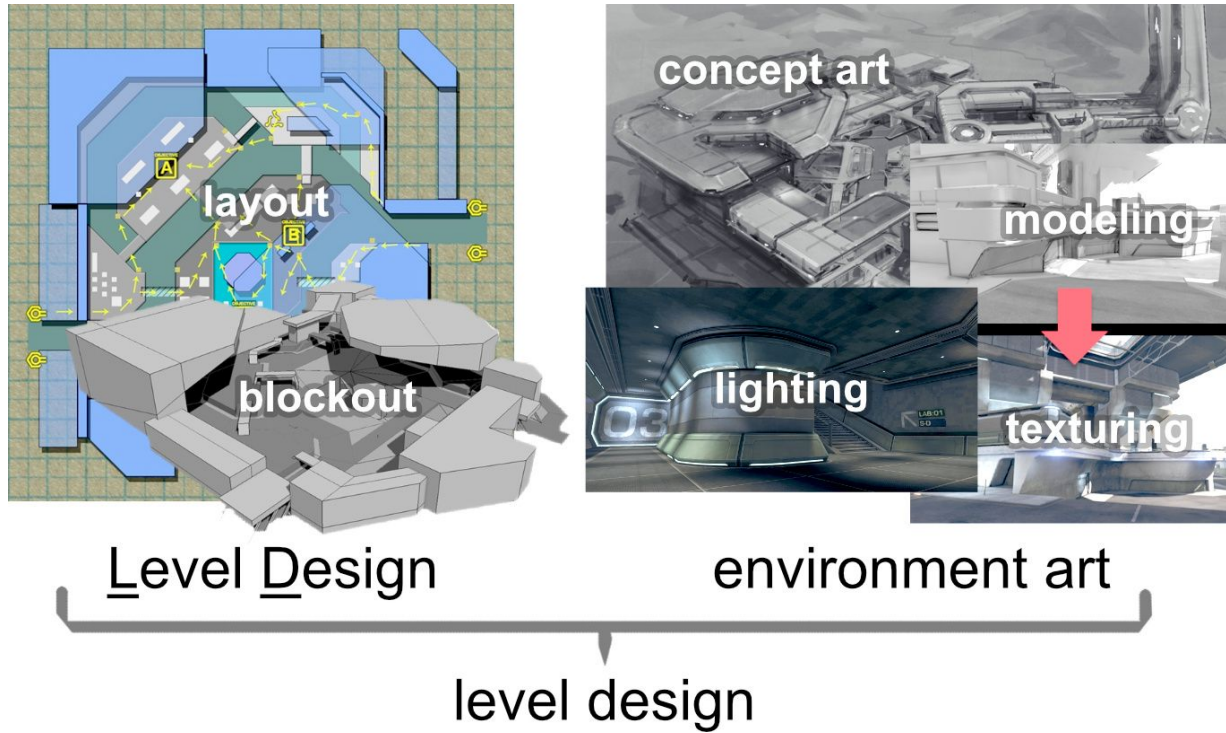
**ANALOGY**

# What is a level?

**Level =**







The Level Design Book: "diagram comparing "Level Design" vs "level design" vs "environment art", using process images of Gallente Research Facility from Dust 514",  
<https://book.leveldesignbook.com/introduction>

**Environment Art**  
(Aesthetics)



**Game Design**  
(Mechanics)

**Level Design**  
(Dynamics)

# Environment Art (Aesthetics)



**Game Design**  
(Mechanics)

**Level Design**  
(Dynamics)



# Environment Art (Aesthetics)



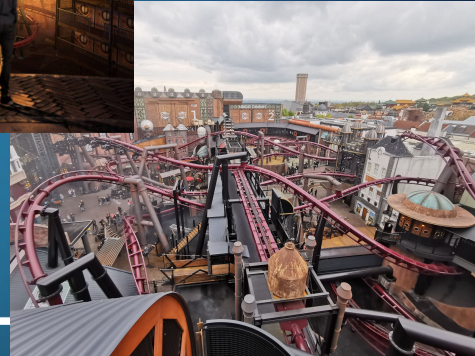
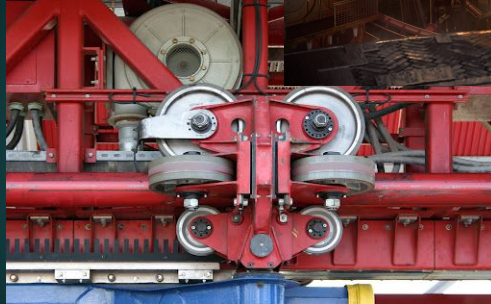
**Game Design**  
(Mechanics)



**Level Design**  
(Dynamics)

Example: Roller coaster F.L.Y. in "Rookburgh",  
Phantasialand, Cologne, Germany

# Environment Art (Aesthetics)



**Game Design**  
(Mechanics)

**Level Design**  
(Dynamics)



# TAKEAWAYS

Level Design  
=  
Design of **Game Spaces +  
Experiences**

(Rules made playable)

# Questions?

# Level Elements.

# So... What does a level consist of?



**Player**

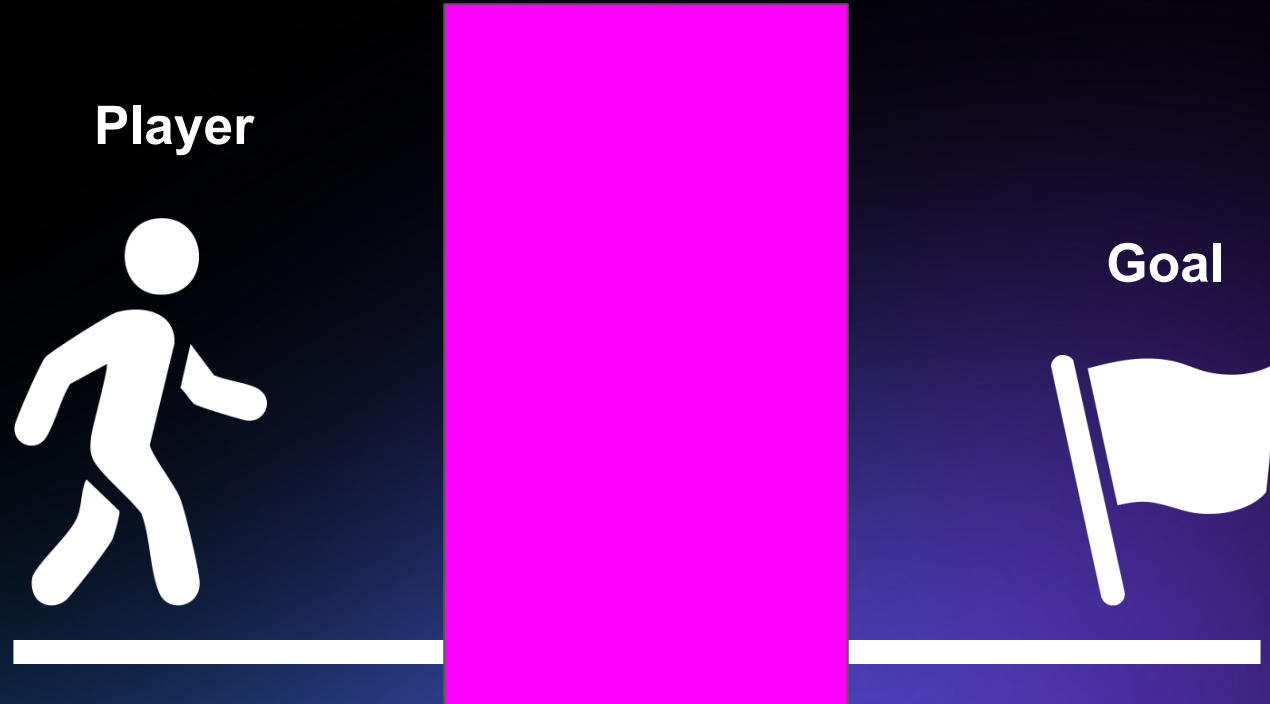


**Goal**

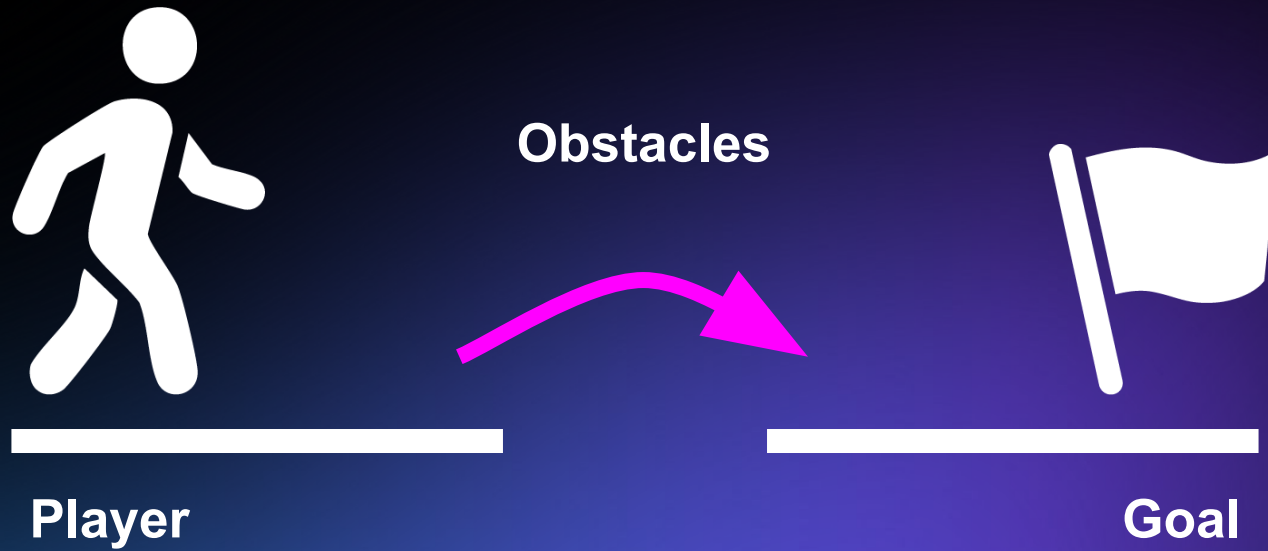
















# Obstacles

# Roadblocks

- **Slows down** without stopping
- Forces to **interact with the level**
- Environmental Obstacles
  - terrain, weather, ...
- Platforming Obstacles
  - platforms, climbing walls, ladders, ...





# Harmful Objects

- Harm players, hinder progress
- **Traps (part of environment)**
  - Spikes, abysses, blades, crush blocks, reward traps, ...
  - should be announced and clearly visible
- **Enemies (non-player characters)**
  - Hostile creatures, enemy soldiers, boss characters, ...



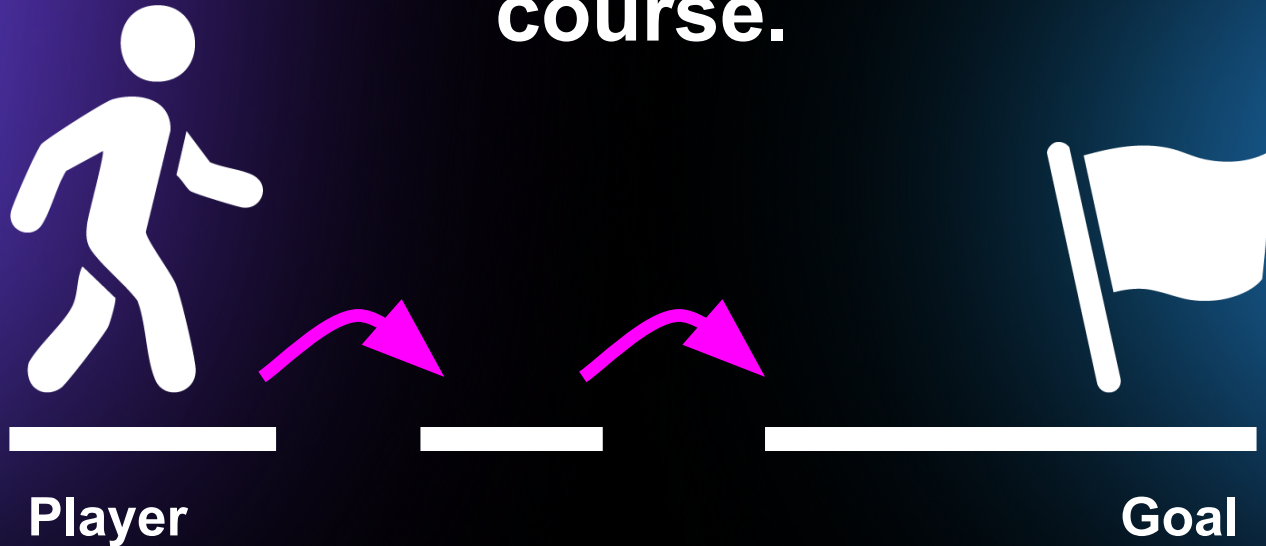
# Puzzle Obstacles (lock & key(s))



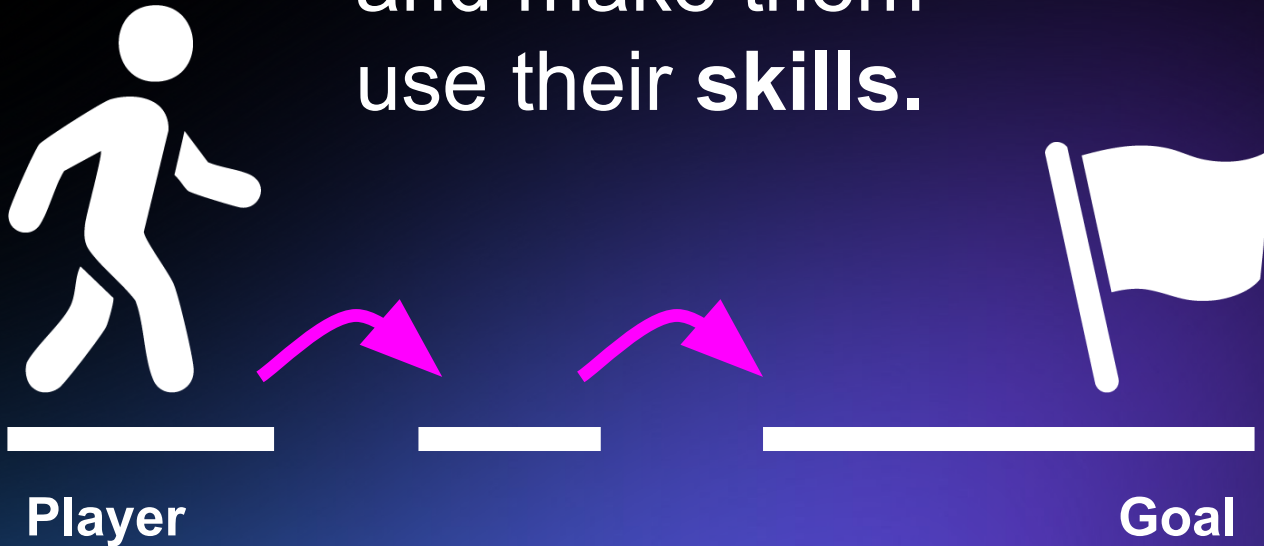
# variety is key!



A level is an  
**obstacle**  
**course.**

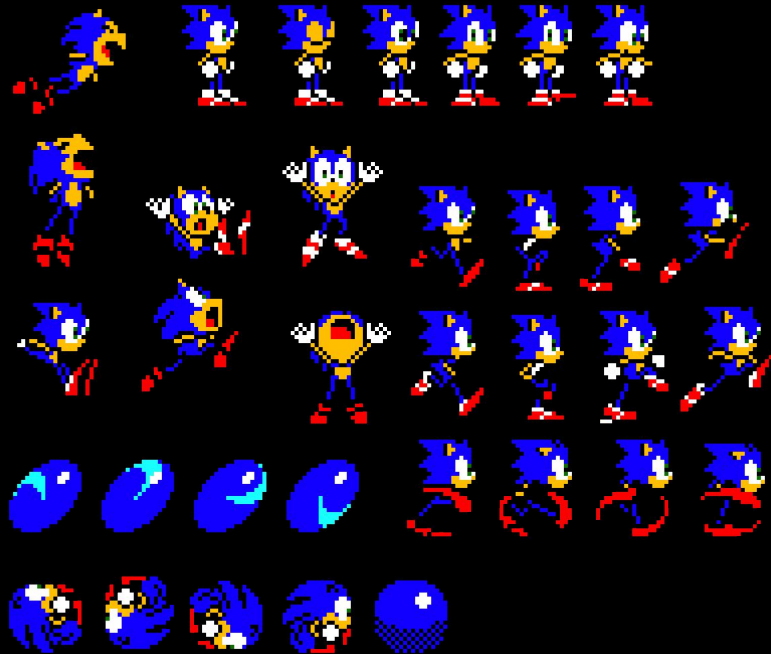


Obstacles  
**challenge** players  
and make them  
use their **skills**.





# Fundamental Skills



Skills are the  
**player's learned abilities**  
**to perceive, interpret, and act**  
within a game's rule system.

# Skills

BACK ○

SELECT ×



**One Skill at a Time**



Skill Points x

0

★  
Defend

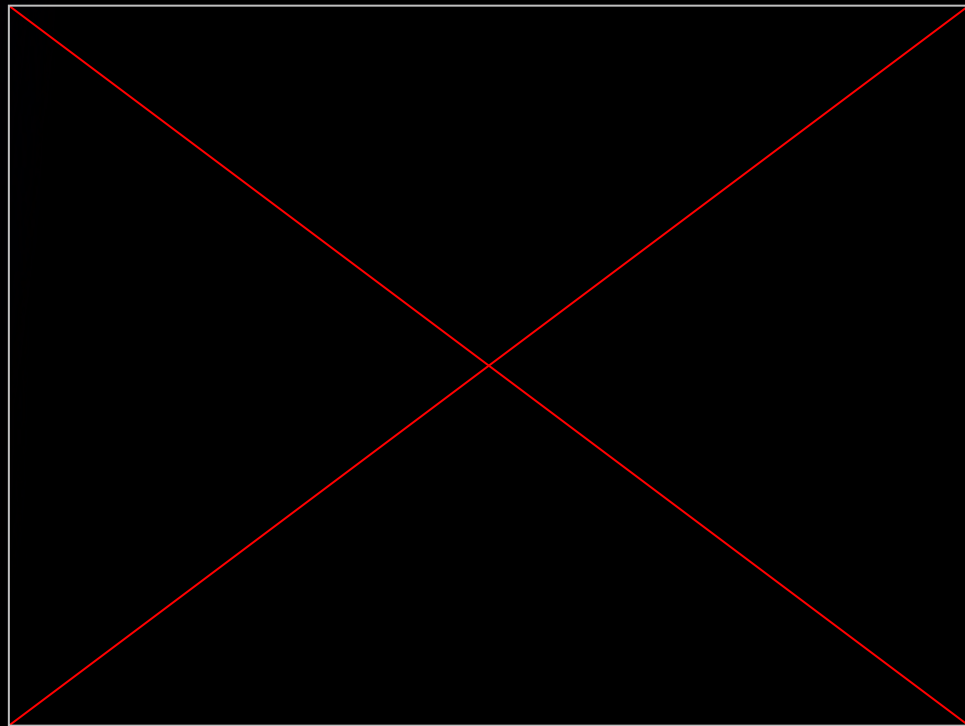
Reduce damage from attacks by 40% and increase speed on the Timeline by 70% for the following turn.

Cast Time: Instant

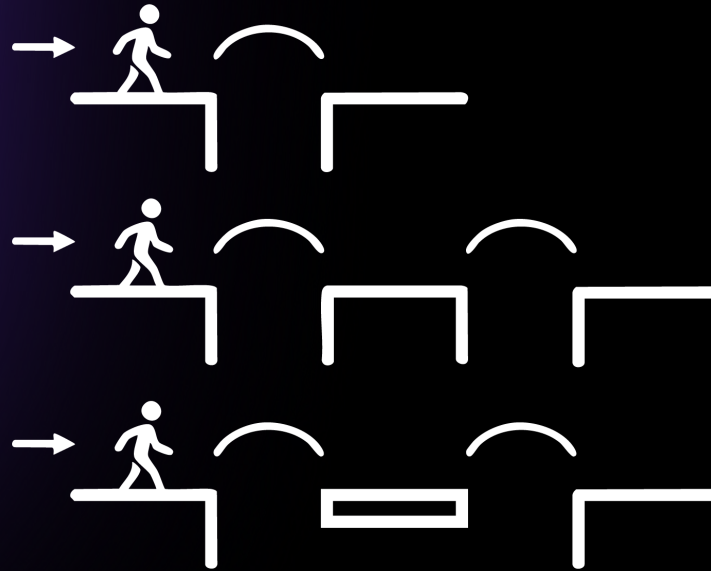
# Introducing new Skills

1. **Safe Introduction** (teaching)
  2. **Practice** (application)
  3. **Test** (mastery)
- should be distributed equally throughout the game

Example: Confront players with an obstacle that requires a new skill.



# Introducing Skills





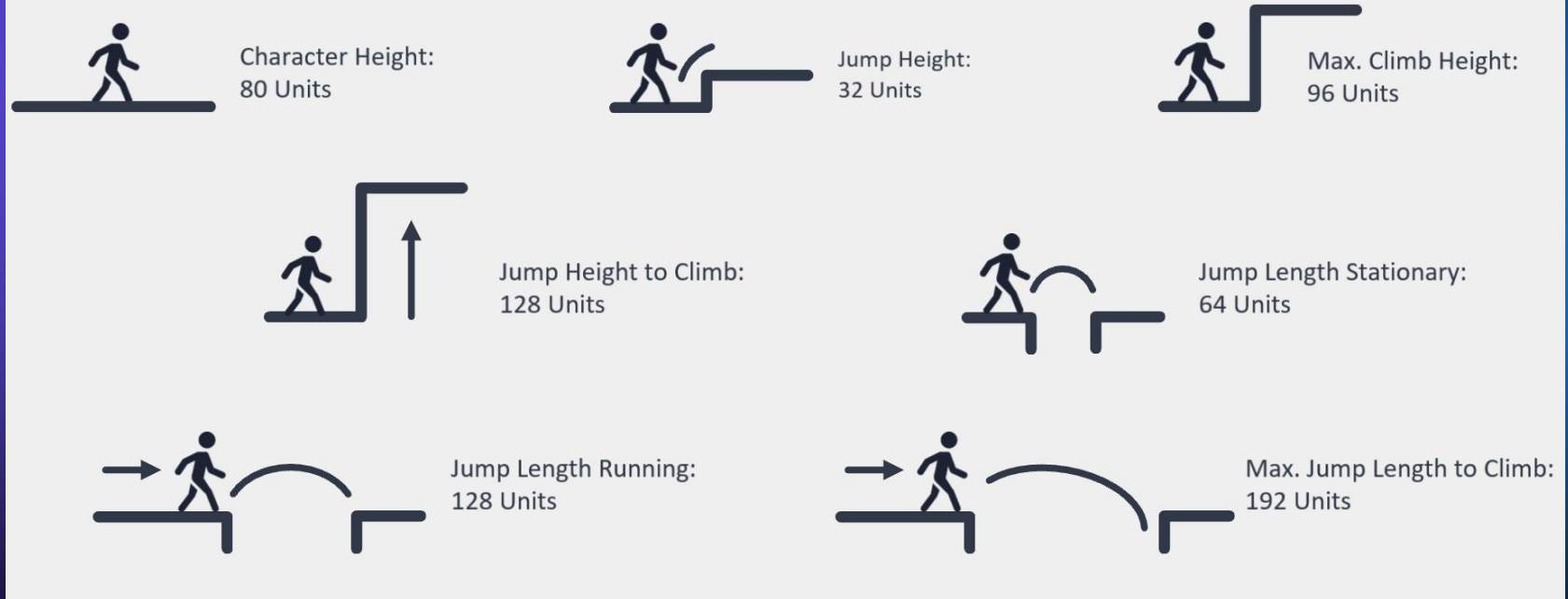
# Skill Gating

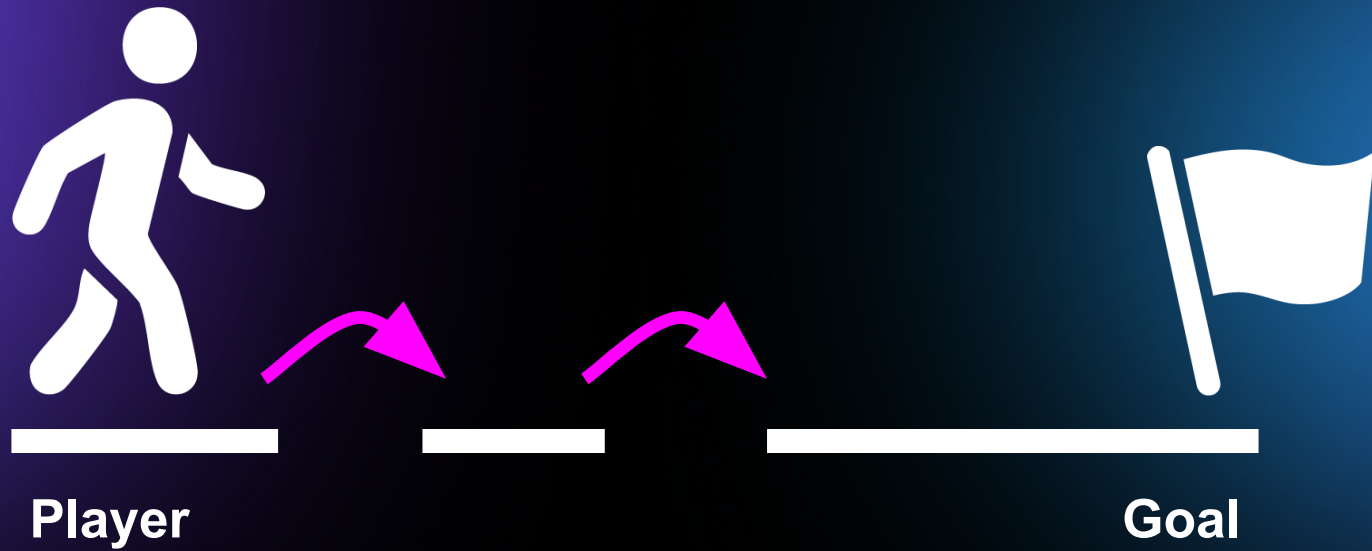
- Hide new ability behind a skill gate
- Progress in the game can be tied to a specific skill of the player (or a skill of the character).
- The player should not become frustrated, but rather understand: “I’m not there yet—but I can get there.”

Example: An abyss that cannot be crossed with a normal jump, but with a double jump



# Combining Skills







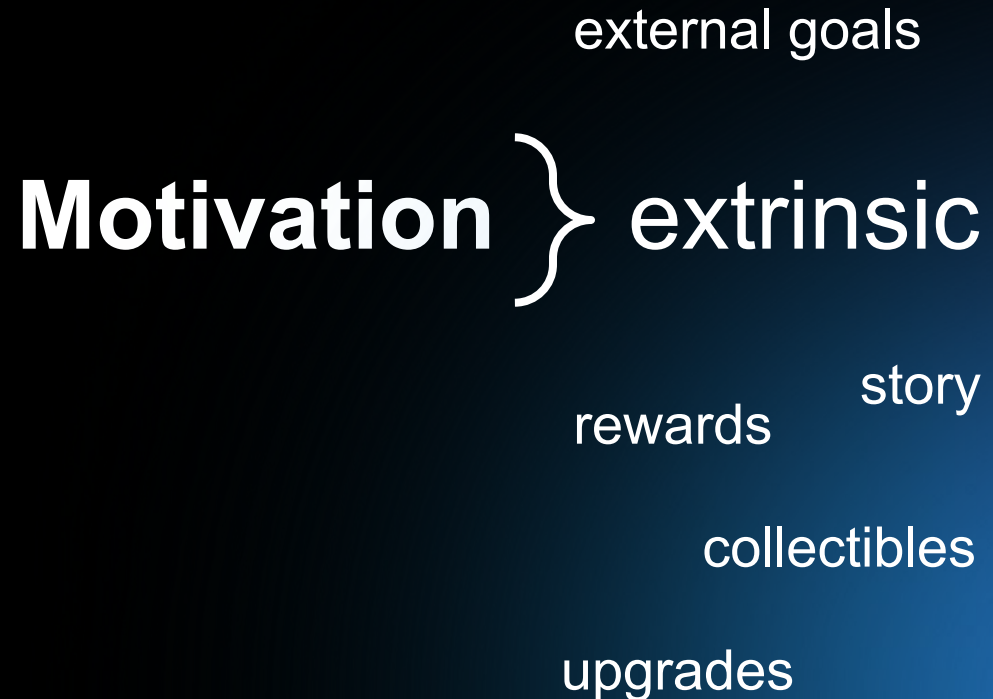
intrinsic { **Motivation** } extrinsic

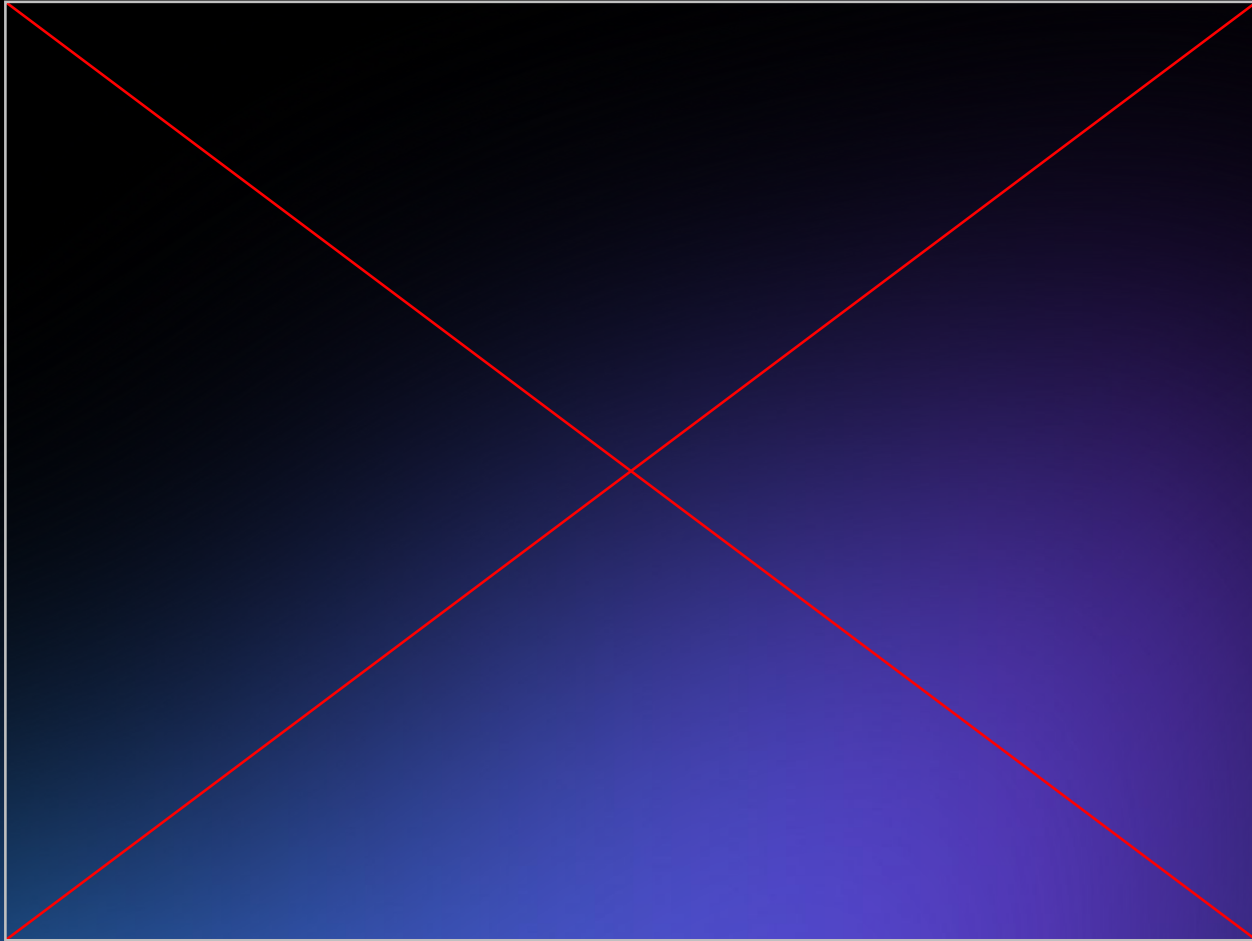


personal enjoyment

intrinsic { **Motivation**

satisfaction in activity





# TAKEAWAYS

**Challenge, skill, and motivation**  
keep space alive — without  
them, levels lose tension and  
meaning.

# Questions?



# Player Experience.

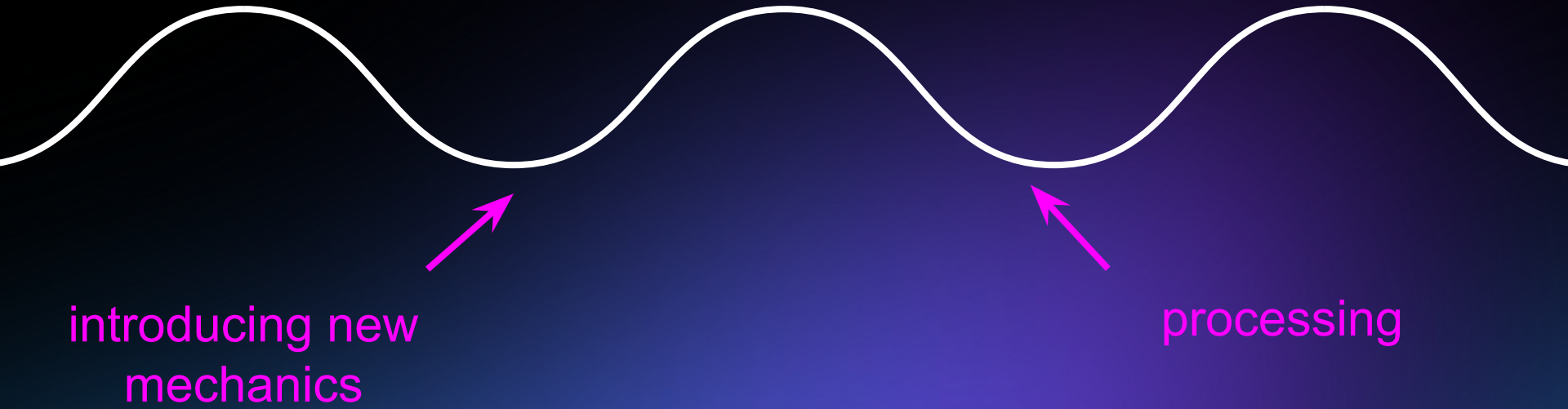
# Pacing

- Tempo and rhythm of a level
- Balance between:
  - tension and relief
  - challenge and reward
- Deliberate use of story, gameplay, music, and visual elements
  - timing of attention





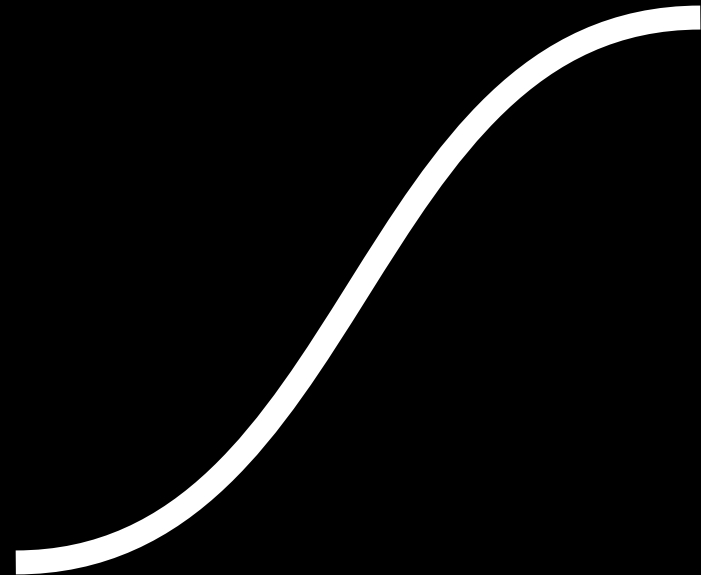
Balanced pacing ensures that players remain motivated and are neither overwhelmed nor underchallenged.



# Difficulty Curve

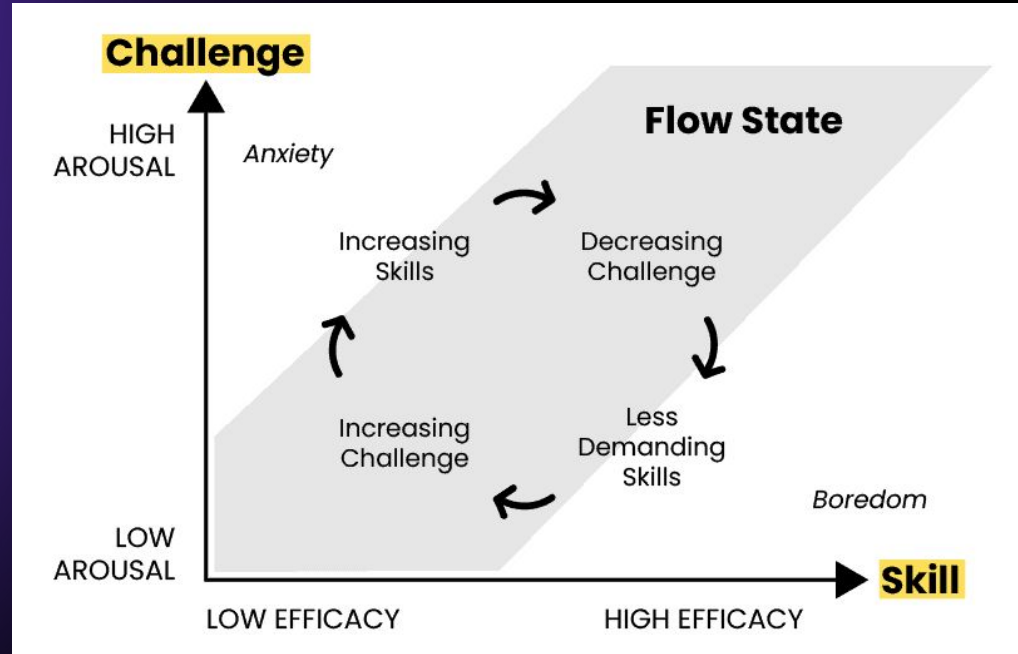
- Exponential curve ->  
Balanced progression
  - steady but understandable  
increase in difficulty and skills
  - without seeming unfair

Increase the intensity not only through variation, but also through combination, tempo, and context changes, not just by adding more opponents.





# Flow



# TAKEAWAYS

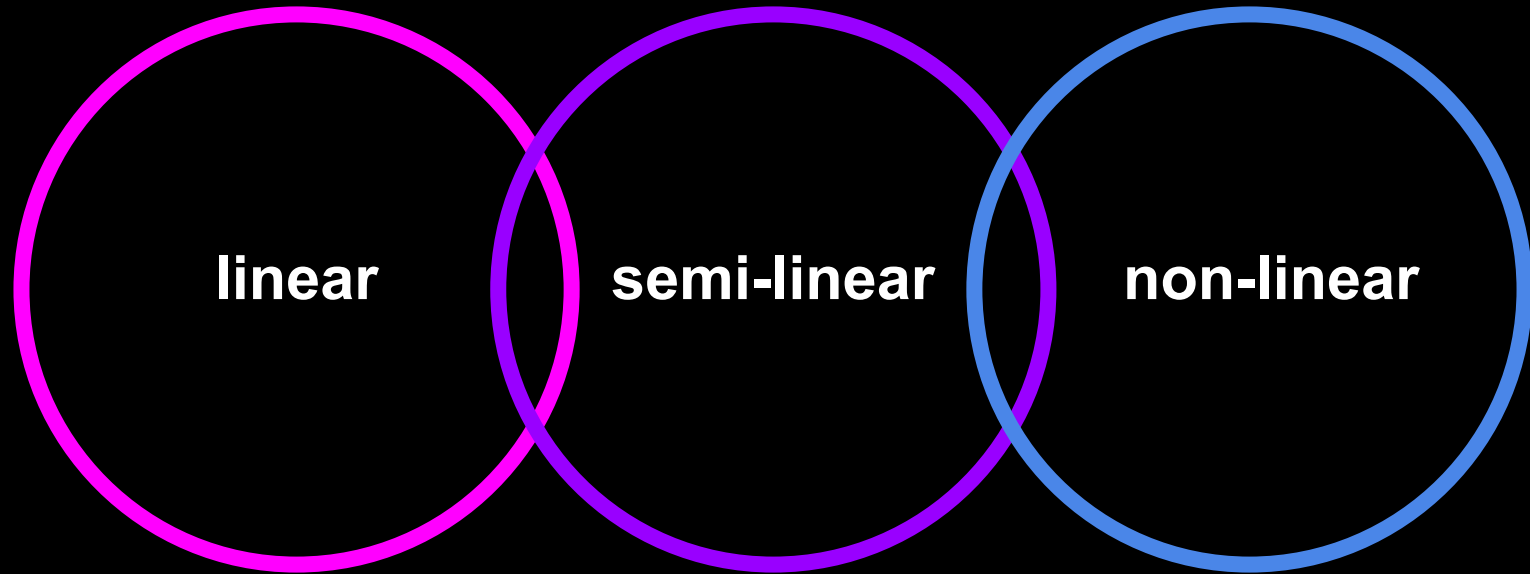
Find the  
**right balance.**

# Questions?

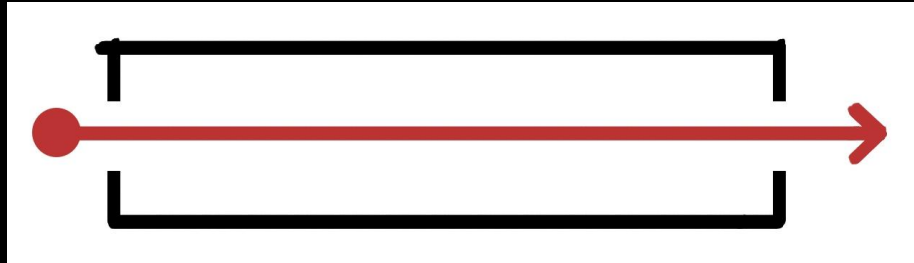
# Spatial Composition.

# Types of Levels.





# Linear Corridor



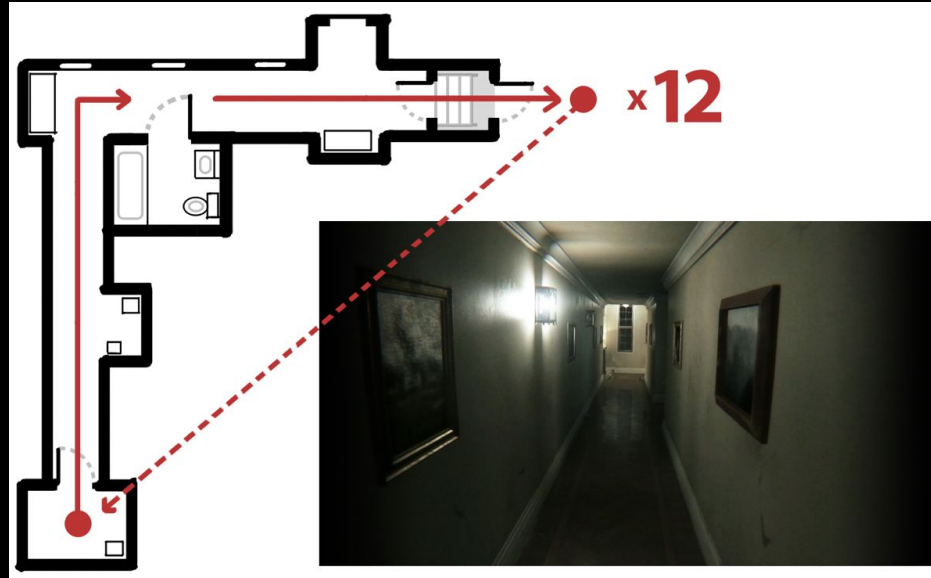
# Linear Corridor



Limbo | Playdead | Playdead | 2010

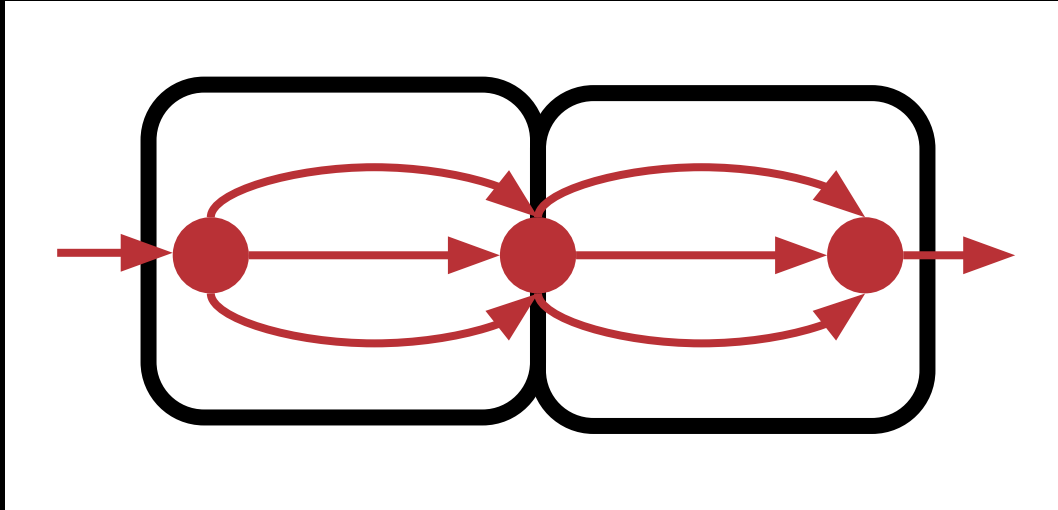
Linear

# L-Shaped Corridor (Loopback)



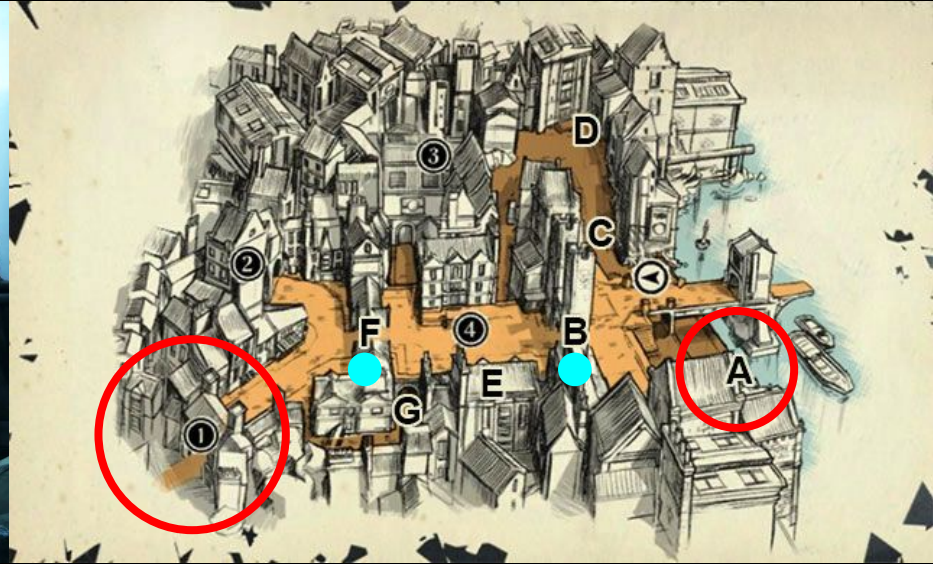
P.T. (Playable Teaser) (Silent Hill) | Kojima Productions | Konami | 2014

## Semi-Linear Branching Levels





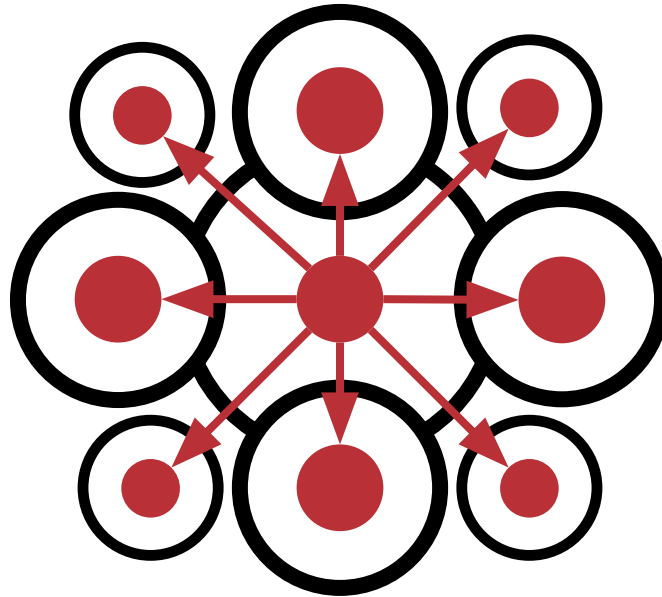
## Semi-Linear Branching Levels



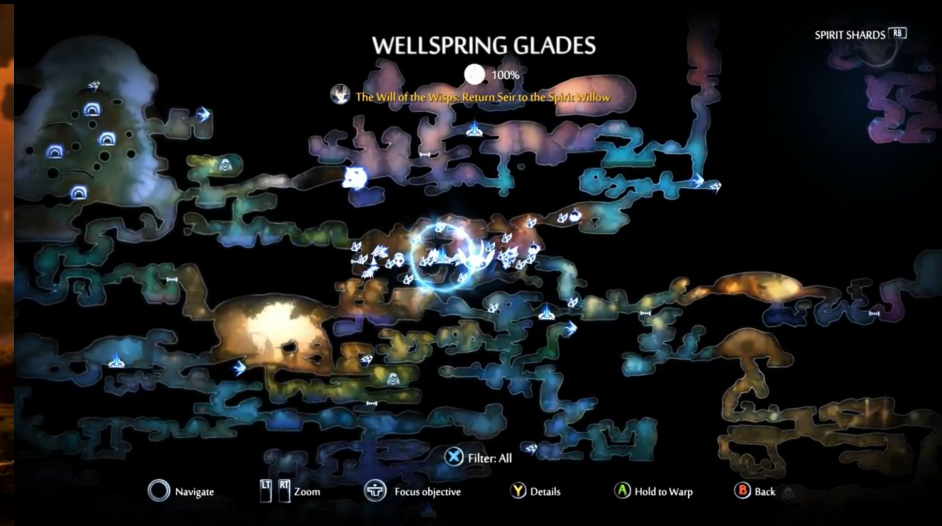
Dishonored | Arkane Studios | Bethesda Softworks | 2012

Distillery District - the Main Part (Mission 2 - High Overseer Campbell)

## Semi-Linear **Hub-and-Spoke**



## Semi-Linear Hub-and-Spoke



Ori and the Will of the Wisps | Moon Studios | Xbox Game Studios | 2020

## Semi-Linear Hub-and-Spoke



League of Legends | Riot Games | Riot Games | 2009-2025 (Arena-Mode)



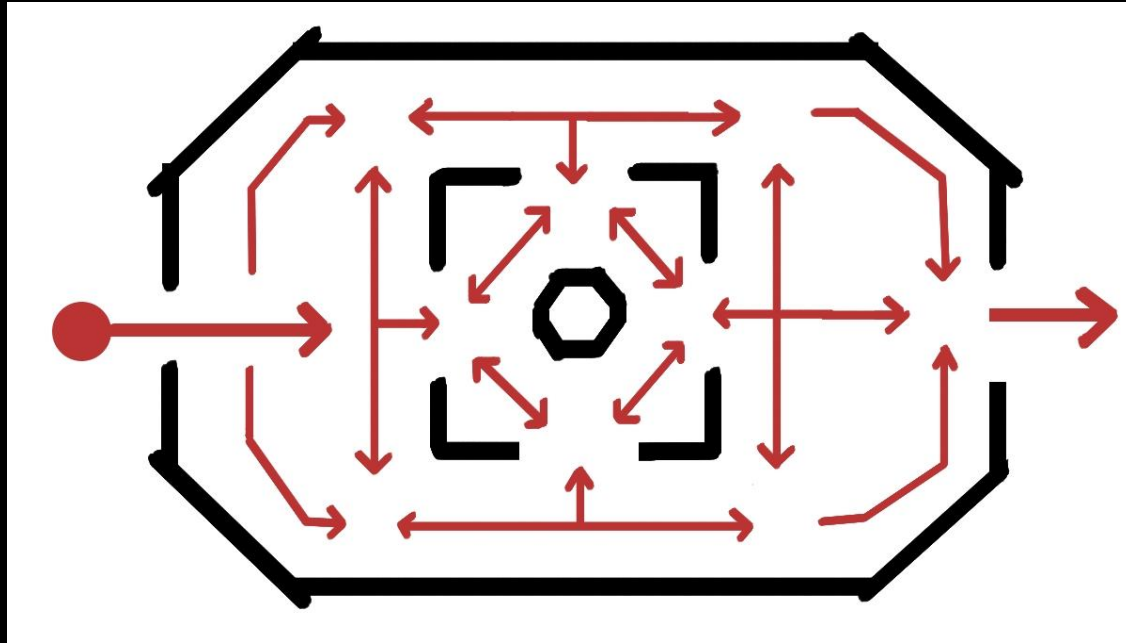
## Semi-Linear Hub-and-Spoke



League of Legends | Riot Games | Riot Games | 2009-2025 (Arena-Mode)

Semi-Linear

# Arena / Point of Interest (POI)





## Semi-Linear

# Arena / Point of Interest (POI)



# Non-Linear Open World

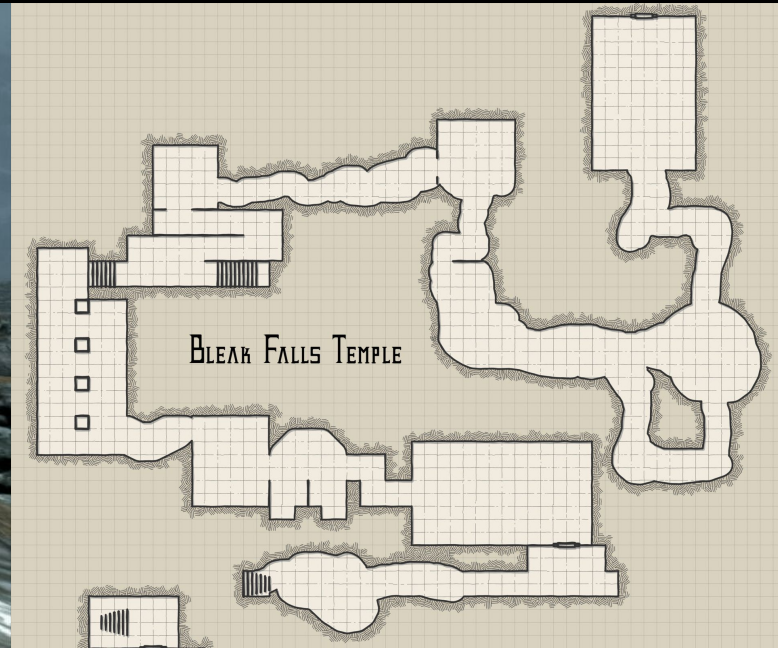


The Elder Scrolls V: Skyrim | Bethesda Game Studios | Bethesda Softworks | 2011



Starbound | Chucklefish | Chucklefish | 2016

# Combinations of Layouts



# Questions?



# Building Vocabulary.

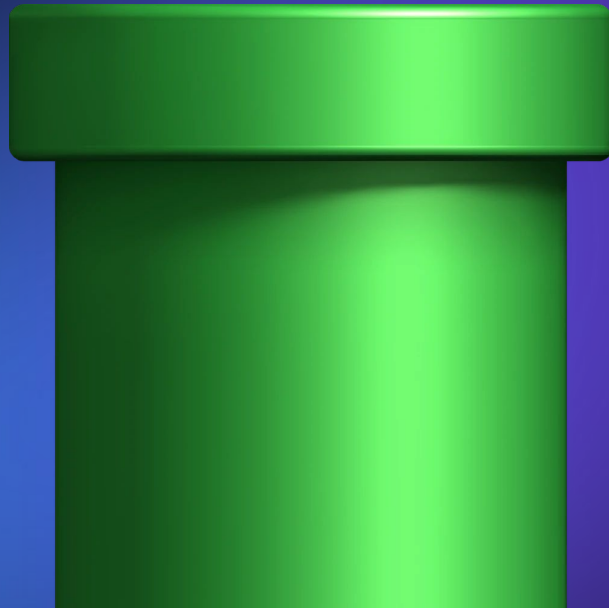




# Affordances

- Perceived call to action of an object or space
- An object “tells” the player what can be done with it
  - through its form, material, context, or experience.

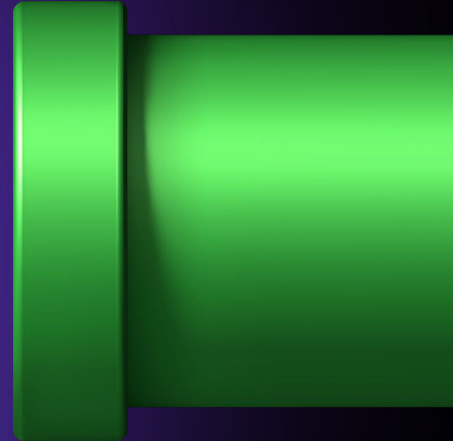




# Readability

**Readability means that the player intuitively understands.**

- The game establishes a visual vocabulary
  - repeats and expands
  - Players learn new meanings with each encounter
- If you don't build or break this vocabulary, cognitive confusion arises.



# Readability Tips

- ☐ **Consistency first**  
(Same form = same function)
- ☐ **Contrast**  
(Interactive objects must be clearly distinguishable from decor)
- ☐ **Hierarchy**  
(Visually prioritize important affordances)
- ☐ **Reinforce through feedback**  
(When the player acts correctly, confirm it (ex. sound))
- ☐ **Seed recognition early**  
(Show central forms early so that it can be recognized later)

# Questions?

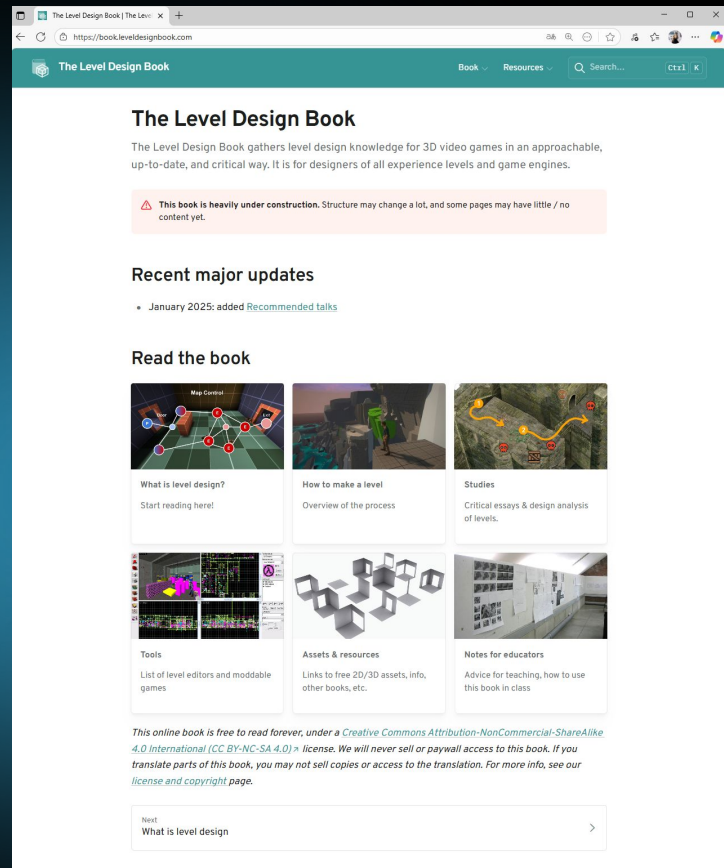
# How to Level Design?



# The Level Design Book

Recommendation: <https://book.leveldesignbook.com/>

- by Robert Yang, Andrew Yoder
- open source, work in progress



# 1. Ideation & Pre-Production

Step 1:  
**Plan the  
big ideas and  
overall experience design.**

# Ideation (W-Questions)

Define/determine/set:

- ☐ **Player Role & Identity**  
(Who is the player in this context?)
- ☐ **Player Experience**  
(What should the player feel and learn?)
- ☐ **Player Goal & Motivation**  
(Why is the player moving through this space?)
- ☐ **Core Mechanics**  
(What is the player allowed to do?)
- ☐ **Setting & Mood**  
(Where does it take place, what atmosphere should it have?)

# Do I need Pre-Production?

- Depends on scope and goals.

There is no universal method for creating a level.

## **Group & Project size?**

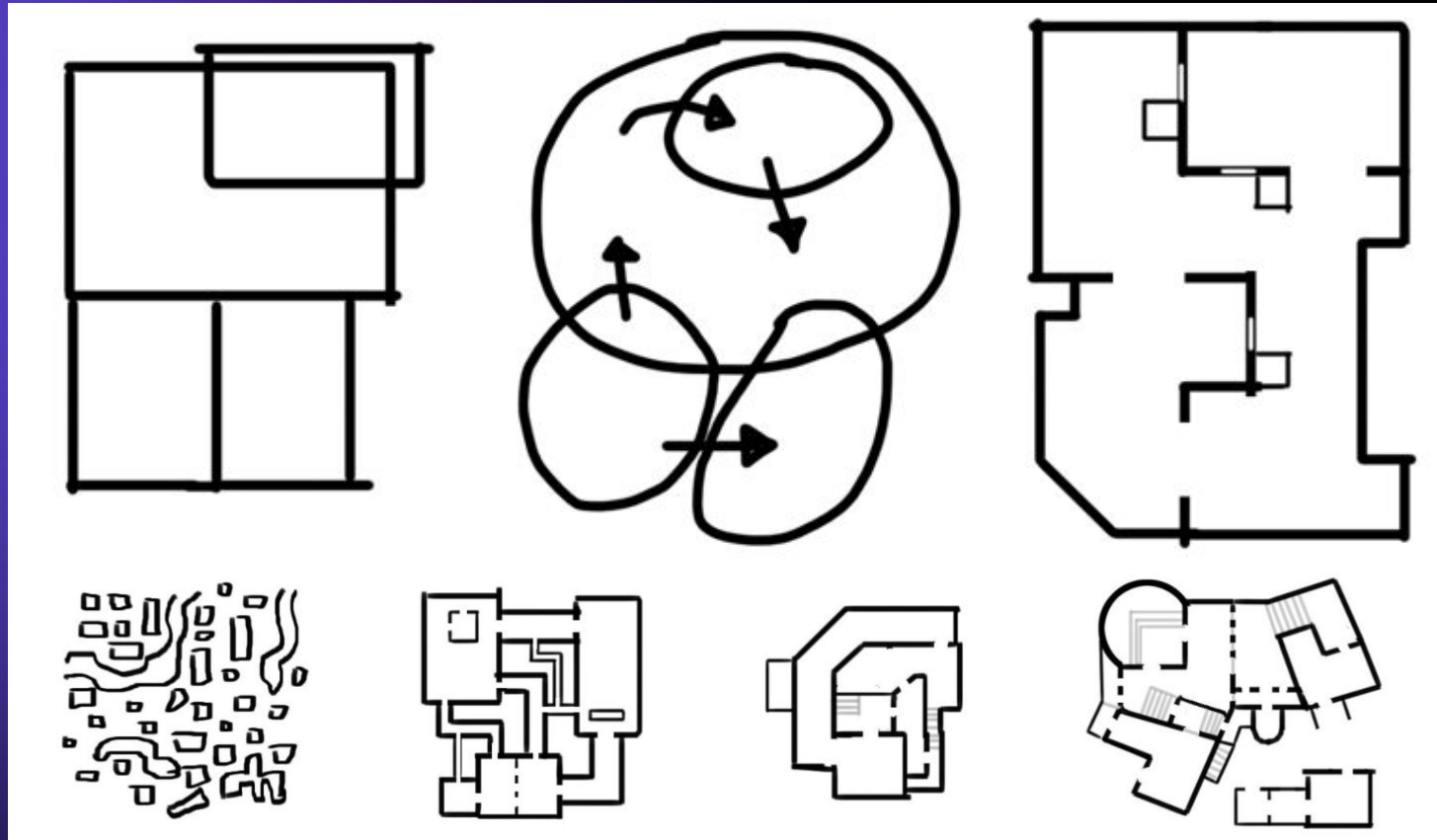
- The larger the project, the more and more precise planning is required.

## **What do I want to make? And how? Genre? Mode of Play?**

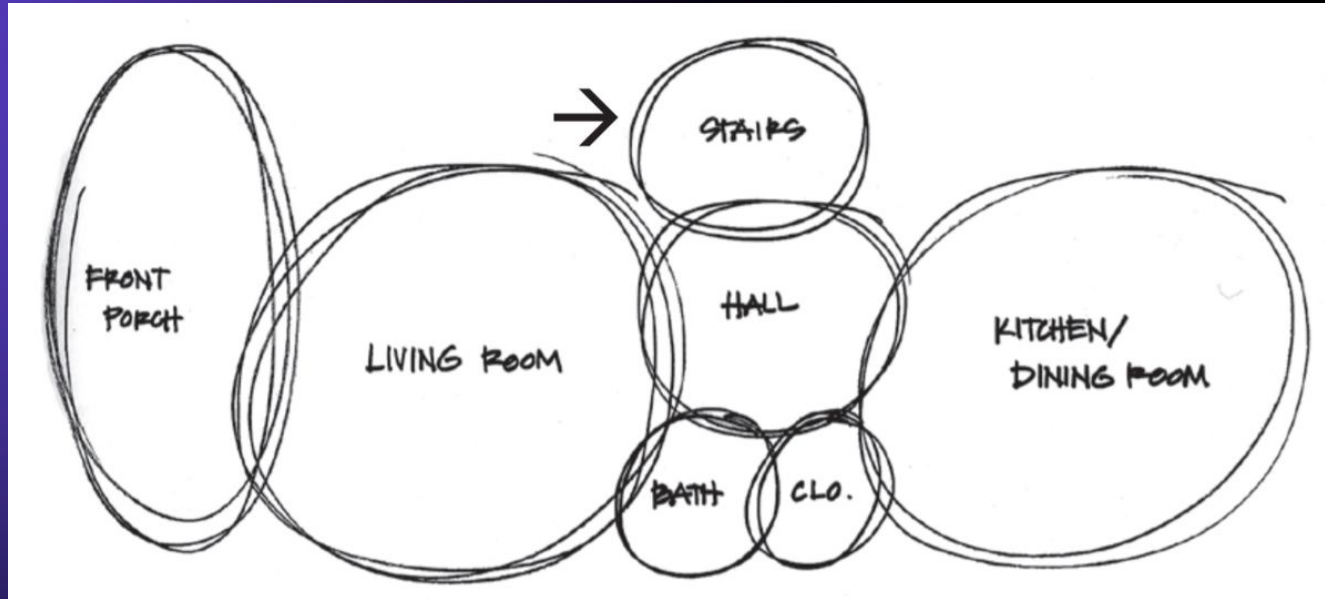
- Multiplayer: needs longer blockout phase → testing balance & encounters.
- Single-player / narrative: focus on pacing & storytelling → shorter blockout phase.

# 2. Layouting

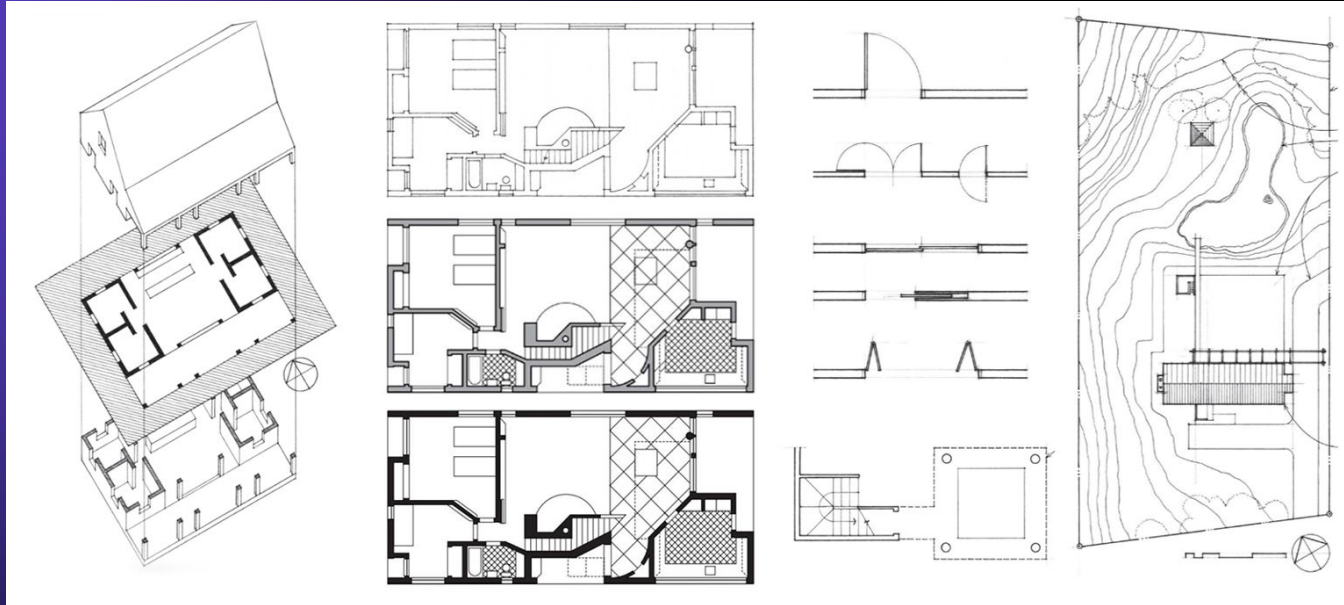




# Bubble Diagram



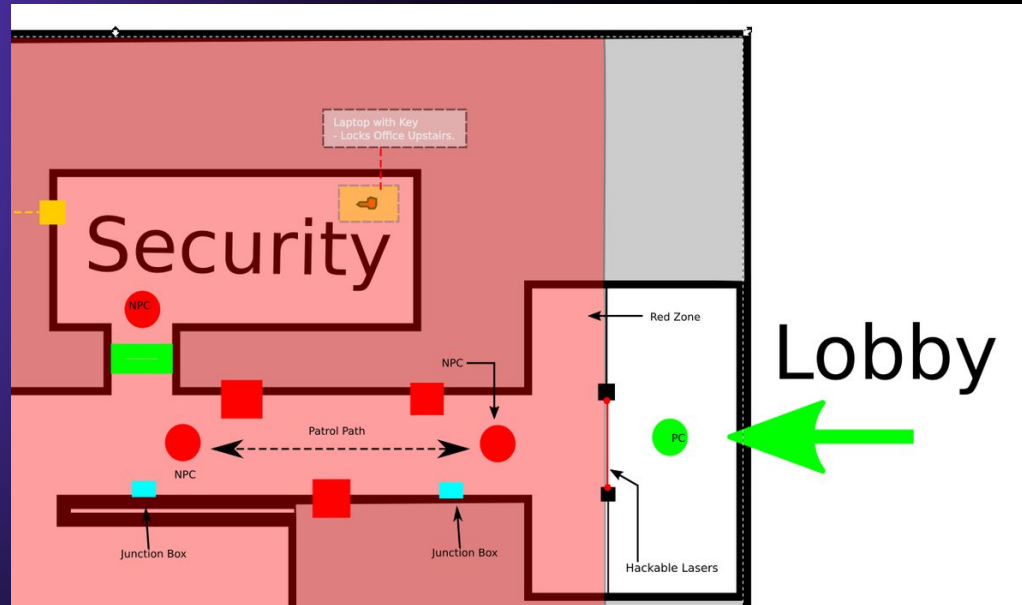
# Floor Plan Drawing



<https://book.leveldesignbook.com/process/layout>,  
"Architectural Graphics" (6th ed.) by Francis D. K. Ching

# Top Down Annotated Map

(+ Markups)

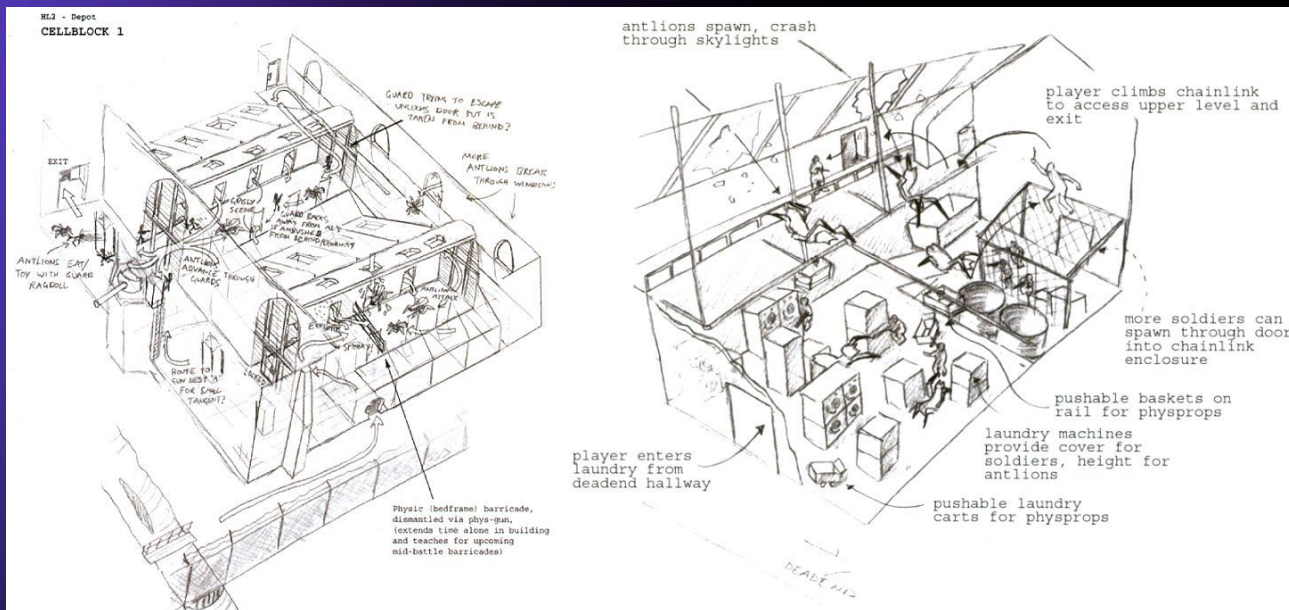


<https://book.leveldesignbook.com/process/layout>,

WKZ Station in "Automata" for Watch Dogs 2, as drawn by the designer Iuliu-Cosmin Oniscu

# Isometric Layouting

## (+ Markups & Player Flow)

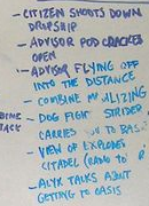


<https://book.leveldesignbook.com/process/layout/>

Nova Prospekt encounters, from art book "Half-Life 2: Raising The Bar"



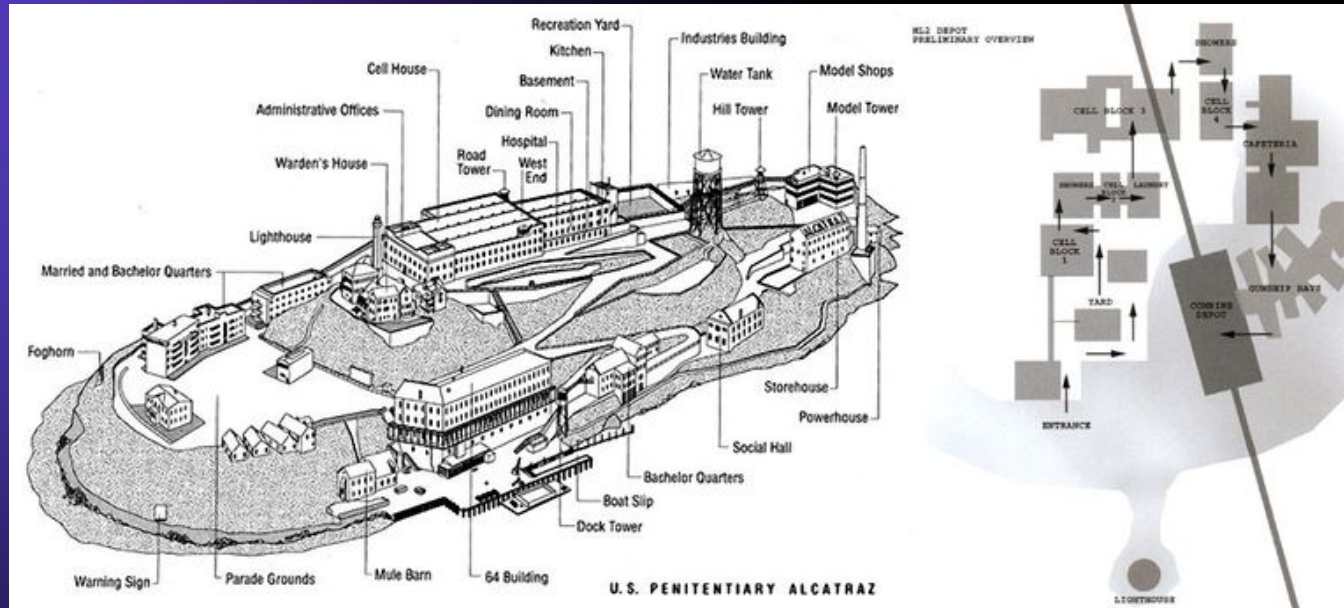
(+ pacing outline)



<https://book.leveldesignbook.com/process/layout>,  
"Phil Co (via [Interlopers.net](https://interlopers.net)), Half-Life 2: Episode  
Two.



# Axonometric Overview + Map

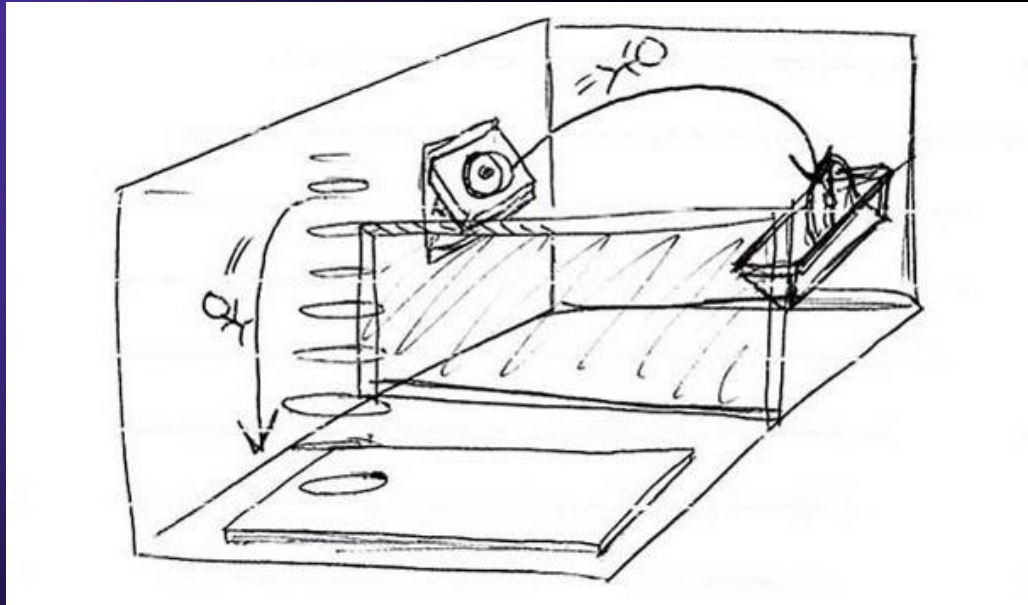


<https://book.leveldesignbook.com/process/layout>

map of Alcatraz (left) and Nova Prospekt cell block bubble diagram (right)  
from art book "Half-Life 2: Raising The Bar"

# Isometric Layout

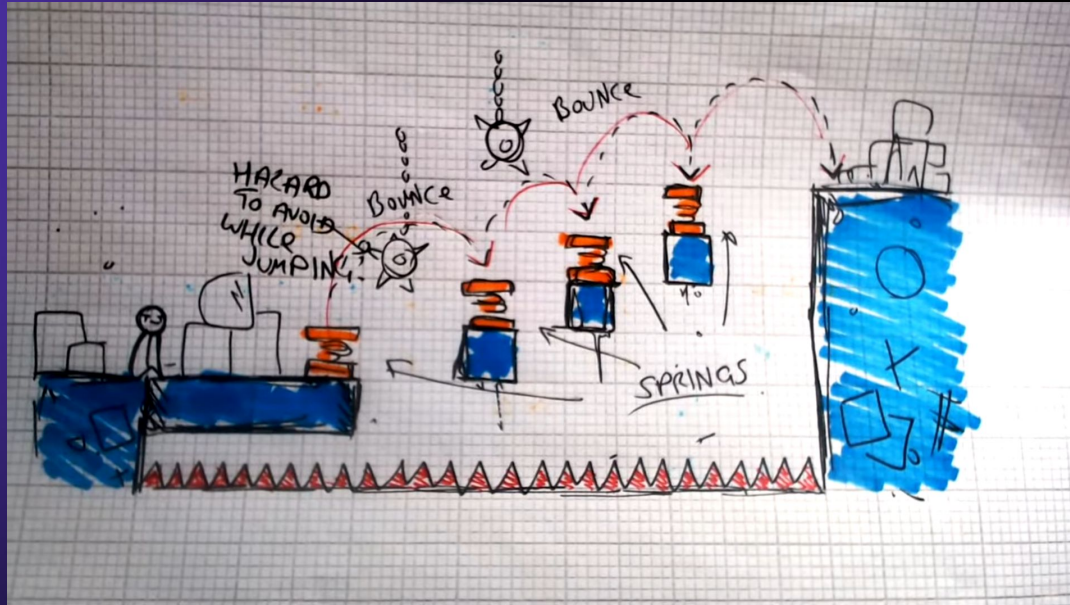
(+ Player Flow Diagram)



<https://book.leveldesignbook.com/process/layout>,  
Portal 2, from Game Informer, March 2010

# Layout Sketch 2D Platformer

(+ Markups & Player Flow Diagram)



<https://www.youtube.com/watch?v=HU25BJH1PrY>  
Let's Make Games: Level Design Tips by Sinclair Strange

[illegible]

105

# UE 5

# UE 5

**Task:** Create a platformer level in Construct.

**Theme:** “Tough but fair” / “Hart aber fair”

**Who:** Each team builds one level.

**How:** Blocked, sprite primitives (no complex graphics), no (minimal) UI.

**What:** Wiki page with information about the level and level for download (Iteration 1).

[hagenberg.games Wiki](#)



# UE 5

## Today's Plan 90 Min

### Block 1: 35 Min (Paper)

1. Ideation (5 Min)
2. Layouting (20 Min)
3. Presenting (10 Min)

### Block 2: 45 Min (Construct 3)

1. Input (15 Min)
2. Blockout & Scripting
3. Testing

-> Next Week: Art Pass

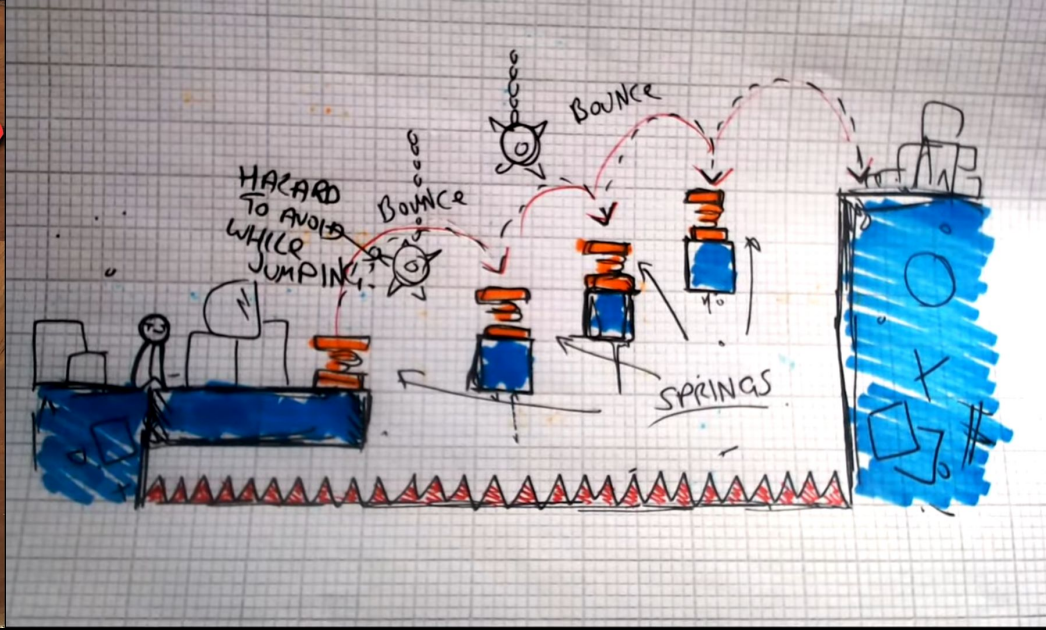
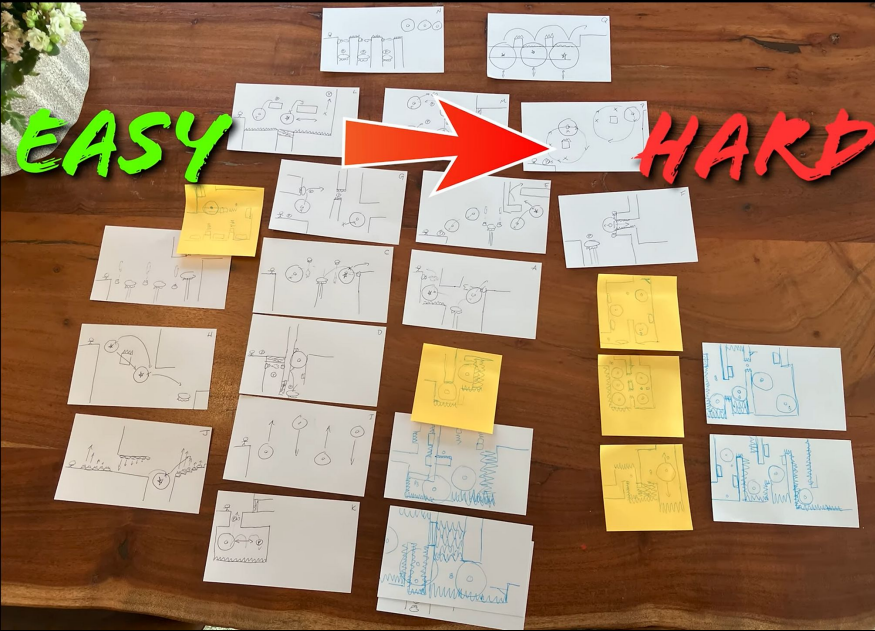
# Ideation (W-Questions)

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- ☐ **Player Goal & Motivation**  
(Why is the player moving through this space?)
- ☐ **Core Mechanics**  
(What is the player allowed to do?)
- ☐ **Setting & Mood**  
(Where does it take place, what atmosphere should it have?)

**TIP**  
Use a random word  
generator  
(noun verb noun)

## 2. Layouting



# 3. Blockout

# Why build a Blockout?

“A blockout (also blockmesh or graybox) is a [...] rough draft level built with **simple** [...] **shapes (primitives)**, but without any details or polished art assets.”

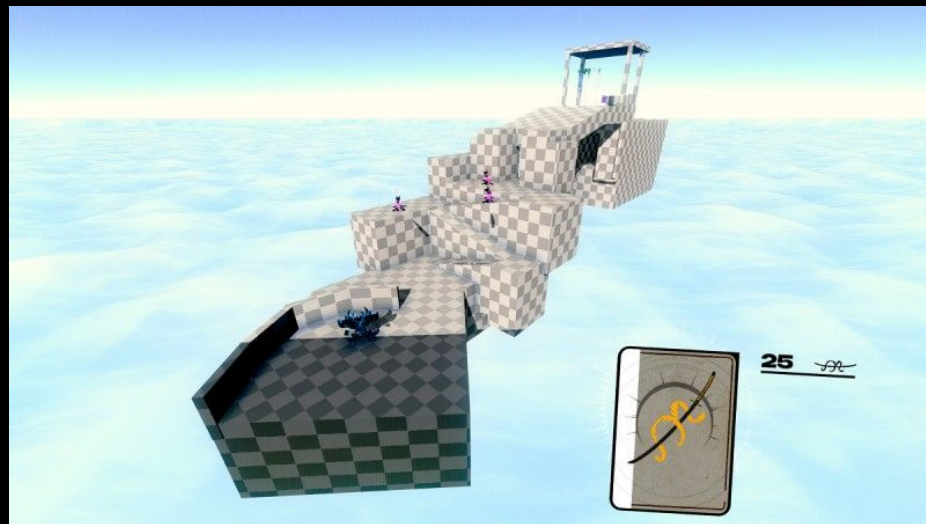
**Goal:** Prototype, test, and adjust the foundational shapes of the level.



[https://book.leveldesignbook.com/process/layout,blockout vs final art-pass for "Docks" in Call of Duty: Modern Warfare \(2019\) by Brian Baker](https://book.leveldesignbook.com/process/layout,blockout%20vs%20final%20art-pass%20for%20Docks%20in%20Call%20of%20Duty%3A%20Modern%20Warfare%20(2019)%20by%20Brian%20Baker)

# How to Blockout:

1. Simple Shapes
2. Functional First (not optical)
3. Greyboxing
4. Readable Silhouettes
5. Iterate Fast



<https://book.leveldesignbook.com/process/layout>,  
early blockout of "Smackdown" in Neon White (2022) by Ben Esposito,  
Russell Honor, and Carter Piccollo (via Game Informer)



# 4. Metrics

# What are Metrics?

“Metrics help you establish scale, the general sense of how big the game world feels.”

- making sure rooms don't feel too small or too big
- makes the level feel more plausible
- build at the player's scale



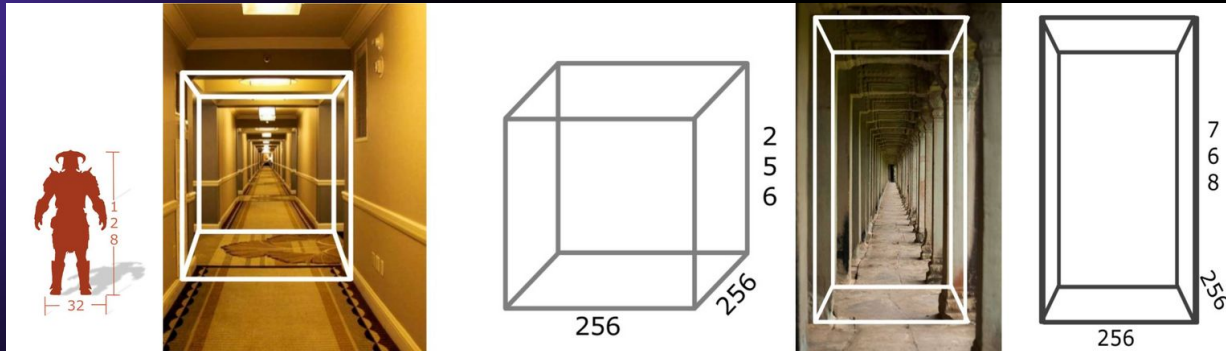
<https://book.leveldesignbook.com/process/layout>, actual internal developer metrics blockout test maps "K\_Test2" and "I\_TestM" from a French version of The Legend of Zelda: The Wind Waker, screenshots by Dark Linkaël (via tcrf.net)

**“Player metrics (or physics metrics) =** factual numbers measured in-engine (player size, movement speed, max. jump height, etc.

- *describes* game physics, derived from the game engine
- example: "player size" is a specific fact that can be measured and verified”

**“Building metrics =** suggested floor / wall dimensions and guidelines that you codify **for yourself based on your desired experience goals**

- *prescribes* game feel, decided by designer(s)
- *example:* "regular hallway size" is a guideline; what does "regular" mean for each project?”





<https://www.youtube.com/watch?v=lcZW-4nrN9I> Min 3:30

How we design 2D platforming levels that are actually fun | Devlog #30 by O. and Co. Games

# 5. Scripting

# What is Scripting?

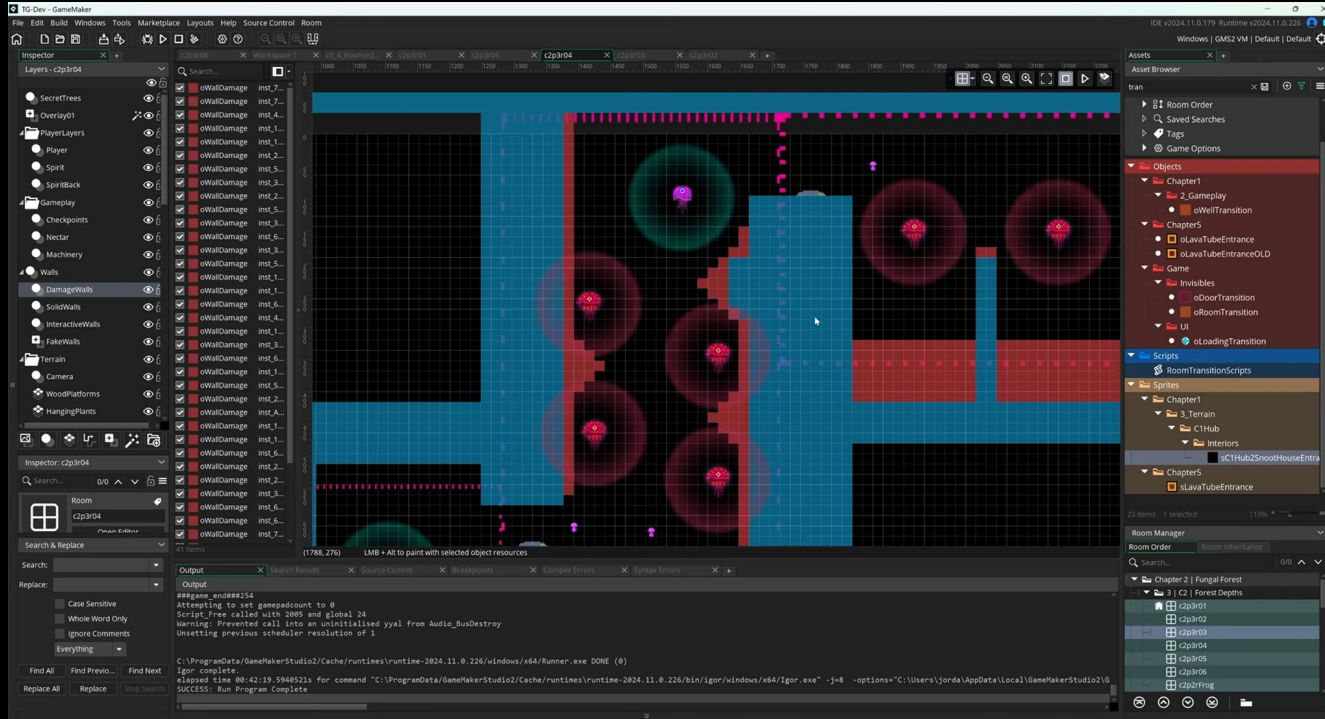
Program specific game rules and behaviors for a level.

Some common examples of level scripting include:

- **Game objects / entities**
  - move or react, like buttons, doors, elevators, traps, etc.
- **Triggers**
  - execute logic when entering an area, to lose health or gain points, etc.
- **Gates block**
  - block progress until the player solves a problem in the game world
- **Objectives**
  - quests, missions



### 3. Blockout / Metrics / Scripting



<https://www.youtube.com/watch?v=lcZW-4nrN9I>

How we design 2D platforming levels that are actually fun | Devlog #30 by O. and Co. Games

# Testing



<https://www.youtube.com/watch?v=lcZW-4nrN9I> Min 4:04

How we design 2D platforming levels that are actually fun | Devlog #30 by O. and Co. Games

# UE 5

**Task:** Create a platformer level in Construct.

**Theme:** "Tough but fair." / "Hart aber fair"

**Who:** Each team builds one level.

**How:** Blocked, sprite primitives (no complex graphics), no (minimal) UI.

**What:** Wiki page with information about the level and level for download (iteration 1).

<https://hagenberg.games/wiki/>

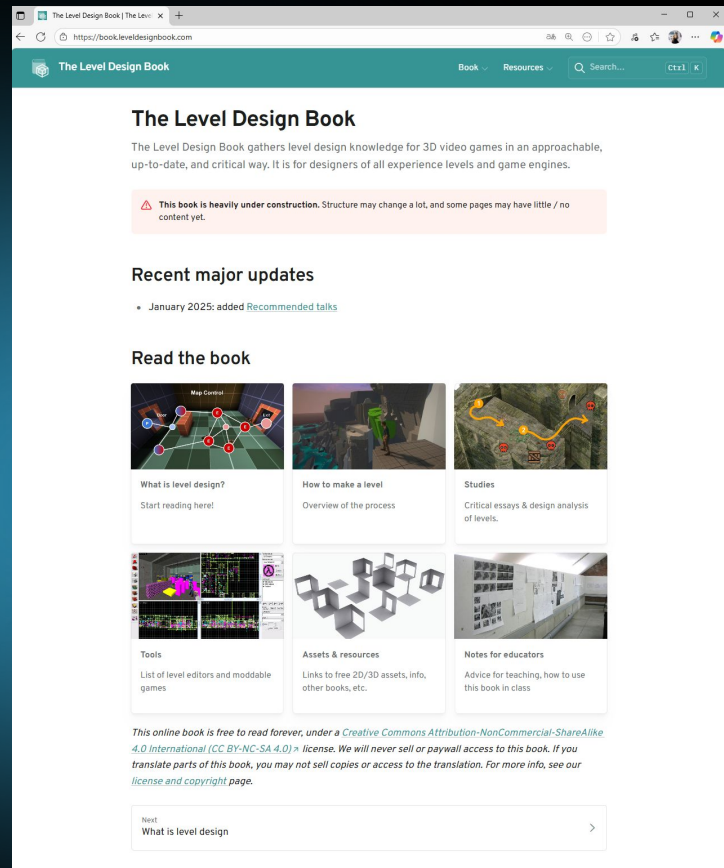
<https://editor.construct.net/>

# Further Tips:

# The Level Design Book

Recommendation: <https://book.leveldesignbook.com/>

- by Robert Yang, Andrew Yoder
- open source, work in progress







<https://www.youtube.com/watch?v=HU25BJH1PfY>  
Let's Make Games: Level Design Tips by Sinclair Strange



<https://www.youtube.com/watch?v=lcZW-4nrN9I>

How we design 2D platforming levels that are actually fun | Devlog #30 by O. and Co. Games

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# Graphics

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